

# Brett Evan Barkley

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## Research Interests

Improving the reliability and scalability of synthetic data through structured augmentation, failure analysis, and generative model diagnostics for more effective learning in both online and supervised settings.

## Education

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| <b>The University of Texas at Austin, PhD</b><br><i>Electrical and Computer Engineering</i> | <b>2022 - Present</b><br><b>GPA: 3.95</b> |
| <b>University of Maryland, M.S</b><br><i>Aerospace Engineering</i>                          | <b>2015 - 2017</b><br><b>GPA: 3.97</b>    |
| <b>University of Maryland, B.S</b><br><i>Aerospace Engineering (Honors Program)</i>         | <b>2010 - 2015</b><br><b>GPA: 4.00</b>    |

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## Selected Research Projects

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| <b>Google Sponsored Research Grant (\$100,000): Scaling Sustainable RL via JAX and Bonsai</b><br>Project Lead                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2025 -      |
| <ul style="list-style-type: none"><li>– Initiated a collaboration with Google, ideated and authored the successful proposal, and am leading the one-year research project.</li><li>– Conducting an empirical study benchmarking JAX against PyTorch in wall-clock speed, cost efficiency, variance reduction, and reproducibility for reinforcement learning and generative modeling.</li><li>– Preparing a publication (NeurIPS, ICLR, or ICML) showcasing how JAX + Bonsai enables faster, more reproducible experimentation with lower computational cost.</li></ul> |             |
| <b>SCOPED: Score–Curvature Out-of-distribution Proximity Evaluator for Diffusion</b><br>First-author Paper Submitted to ICLR 2026                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025        |
| <ul style="list-style-type: none"><li>– Developed a novel framework for OOD detection that reduces inference cost by order of magnitude compared to prior diffusion-based methods. <a href="#">Paper</a>.</li><li>– Demonstrated strong generalization across vision and reinforcement learning benchmarks without retraining.</li><li>– Designed for deployment at scale, with fully parallelizable probes and low overhead suitable for real-time or compute-limited settings.</li></ul>                                                                              |             |
| <b>Fixing That Free Lunch: When, Where, and Why Synthetic Data Fails in Model-Based Policy Optimization</b><br>First-author Paper Submitted to ICLR 2026                                                                                                                                                                                                                                                                                                                                                                                                                | 2025        |
| <ul style="list-style-type: none"><li>– Developed a novel framework (FTFL) to diagnose and fix synthetic data failures in Model-Based Policy Optimization (MBPO). <a href="#">Paper</a>.</li><li>– Identified two systemic issues — output–target scale mismatch and residual prediction variance inflation — and introduced lightweight remedies.</li><li>– Restored performance where MBPO previously collapsed, surpassing model-free baselines (SAC) in 5 of 7 DeepMind Control Suite tasks while maintaining strong results in OpenAI Gym.</li></ul>               |             |
| <b>Stealing That Free Lunch: Exposing the Limits of Dyna-Style Model-Based Reinforcement Learning</b><br>First-author ICML 2025 Paper                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2024 - 2025 |
| <ul style="list-style-type: none"><li>– Developed a JAX-based training pipeline yielding up to a 40× speedup in wall-clock time for MBPO. <a href="#">Code</a>. <a href="#">Paper</a>.</li><li>– Showed that model-based RL algorithms like MBPO exhibit strong performance in OpenAI Gym but often fail in DeepMind Control Suite tasks when trained from scratch.</li><li>– Investigated several potential explanations for this discrepancy, including model error and mitigation strategies; found that even modern techniques fail to close the gap.</li></ul>     |             |

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## Professional Experience

**Autonomy Aerospace Engineer**, Johns Hopkins University Applied Physics Lab (JHU/APL)

2017 - 2022

- Efforts culminated in first ever combat tests between AI and human-piloted F-16s [in 2023](#)
  - JHU/APL's Air Combat Evolution ([ACE](#)) deep reinforcement learning (DRL) lead for sub and full-scale aircraft
  - Guidance, control, and aerospace simulation subject matter expert (SME) for JHU/APL [ADT](#) and ACE teams
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## Technical Skills

**Languages:** Python, C++, Cython, Bash, CUDA

**Libraries/Software:** JAX, Pytorch, Flax, Brax, Git,  $\LaTeX$

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## Selected Publications

1. [Fixing That Free Lunch: When, Where, and Why Synthetic Data Fails in Model-Based Policy Optimization](#)  
**Brett Barkley**, David Fridovich-Keil | *In submission, ICLR 2026*
2. [SCOPED: Score–Curvature Out-of-distribution Proximity Evaluator for Diffusion](#)  
**Brett Barkley**, Preston Culbertson, David Fridovich-Keil | *In submission, ICLR 2026*
3. [Stealing That Free Lunch: Exposing the Limits of Dyna-Style Reinforcement Learning](#)  
**Brett Barkley**, David Fridovich-Keil | ICML 2025
4. [An Investigation of Time Reversal Symmetry in Reinforcement Learning](#)  
**Brett Barkley**, Amy Zhang, David Fridovich-Keil | L4DC 2024